

Talk Like a Structured Graph: Augmenting Text Encodings with Structural Statistics

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Abstract

Does stating a graph’s structural statistics help a language model answer queries about a graph written as text? We prepend each node’s degree, local clustering or random-walk return probabilities to GraphQA’s incident encoding and compare the same graphs with and without them, in paired tests at several edge densities, for Qwen3-1.7B and Qwen3-4B with and without thinking, controlling for answer disclosure, added text and truncation. Stating the answer helps models that compute it unreliably (+7.5 and +8.8 points on node degree) and costs one that computes it reliably (−6.5), and the same values in a longer primer can cost accuracy. Other statistics have smaller effects on average, whose sign depends on the statistic and the model; where structure-free text lowers accuracy, clustering and RWSE mostly do not, and they cut the non-edges plain Qwen3-1.7B states as edges. On cycle detection only about a third of the correct answers given without thinking or a primer rest on a real cycle, against 80.9–96.4% with thinking.

1 Introduction

Research question. When a black-box language model reads a textual graph, do explicit structural statistics improve execution of graph queries? By *execution* we mean computing the answer from the graph’s edges.

Graph models benefit from structural encodings, and how a graph is written as text changes an LLM’s accuracy on graph queries (§2); whether a frozen LLM can use such statistics when they are written as sentences is open. A statistic can help in three ways: it can *state* the answer, change the *procedure* by which the model answers, or add *side information* for reading the graph.

Contributions. (i) Paired tests of three structural statistics and their combination on GraphQA’s six tasks, for four model arms at up to seven edge

densities; (ii) controls that separate what a primer states from what it adds—a graph-blind solver, a structure-free filler and truncation as its own outcome; (iii) evidence that a stated answer substitutes for execution rather than improving it, while thinking does improve it; (iv) checks showing that many correct answers rest on structure the graph lacks.

2 Related Work

Graphs as text. LLM accuracy on graph queries depends on the text encoding, the task and the graph (Fatemi et al., 2024; Wang et al., 2023), and falls as graphs and computations grow (Tang et al., 2025; Jin et al., 2024). Theory separates retrieval tasks such as node degree, which a single attention layer can solve, from connectivity, whose required depth grows with the graph (Sanford et al., 2024). Trained alternatives learn a graph encoding as a soft prompt (Perozzi et al., 2024) or instruction-tune on graph problems (Chen et al., 2024); we change only the text. Structural descriptions need not be read as topology (Huang et al., 2024).

Encodings, controls and faithfulness. Degree and random-walk encodings help trained graph models (Dwivedi et al., 2022; Rampásek et al., 2022). Partial-input baselines expose answers a dataset gives away (Feng et al., 2019); a description’s modality and order, and added, irrelevant or reformatted text, change LLM answers (Das et al., 2024; Ge et al., 2025; Levy et al., 2024; Shi et al., 2023; Liu et al., 2024; Sclar et al., 2024); and a correct answer need not come from the computation a model states (Turpin et al., 2023; Dziri et al., 2023).

3 Method

Design. We use GraphQA’s tasks, its incident encoding (the most accurate for degree and neighbor queries) and its accuracy scoring. The statistics are the graph-learning encodings—node degree, lo-

Task	Arm	None	Degree	Cluster.	RWSE	All	Comp.	Filler
Degree	1.7P	60.50	+7.5	+3.0	+4.0	+10.2	+1.8	-0.5
	1.7T	76.25	+8.8	+1.5	+2.5	+12.2	-1.0	-3.0
	4P	99.25	-6.5	-0.5	-2.2	-8.0	+0.2	-0.5
	4T	97.00	+2.5	+1.8	-1.2	+0.5	+0.2	+2.0
Neighbors	1.7P	72.50	+3.5	+1.2	+1.0	+2.0	+0.8	-6.2
	1.7T	78.75	+6.2	+2.0	+0.2	+6.2	+2.5	+2.5
	4P	86.00	-3.0	+1.0	+0.0	-2.8	+9.2	-12.5
	4T	95.50	+0.5	-0.8	+0.5	-1.2	+0.5	+1.2
Edge count	1.7P	1.25	+3.0 [†]	-0.5 [†]	-0.8 [†]	+0.8 [†]	-1.0 [†]	-0.5 [†]
	1.7T	15.75	-3.2 [†]	-1.2 [†]	-3.2 [†]	+5.0 [†]	-0.8 [†]	-1.5 [†]
	4P	2.00	+27.2	-1.2	-1.5	+3.8	-1.2	+1.2
	4T	20.00	+8.5[†]	+24.8[†]	+10.2[†]	+1.8[†]	+24.8[†]	+12.2[†]
Edge exists	1.7P	70.50	+4.0	+3.8	+5.0	+11.5	-8.2	-15.8
	1.7T	99.00	-2.5	-1.0	-1.2	-1.8	-0.8	-0.5
	4P	90.50	-4.8	+4.0	-10.0	-10.0	-2.8	-7.0
	4T	99.75	+0.0	-0.8	-1.2	-1.0	-0.5	+0.0
Node count	1.7P	31.50	+6.8	+67.2	+2.0	+47.0	+21.2	+39.8
	1.7T	91.00	+5.5	+9.0	+9.0	+9.0	+1.5	+9.0
	4P	90.25	+8.0	+8.8	+9.2	+9.8	+9.0	+9.2
	4T	99.50	+0.2	+0.2	+0.2	+0.5	+0.0	+0.5
Cycle	1.7P	90.75	-17.5[†]	+8.5	-0.2	+7.0	+2.0	+3.0
	1.7T	97.00	-21.8[†]	-0.5	+2.2	-0.8	-7.5	+3.0
	4P	99.00	-2.2	+0.5	-0.2	-0.2	-2.0	-0.5
	4T	99.25	-5.5	+0.0	+0.2	-0.5	-3.2	+0.5

Table 1: Main-sweep correct shares: no-primer baseline (% of all prompts) and paired change (points) on the same graphs at four densities ($p \leq .50$). Bold: $q < .05$ within (model, task, control); [†]: at least 15% truncation on either side; shaded: the primer states the answer. Neighbor sets: exact set match (set-F1 in Table 5(a)); edge existence: raw correctness (balanced accuracy in Table 5(b)). Node count and cycle check have a constant answer at 40 nodes; truncation shares are in Table 6. P/T: plain/thinking.

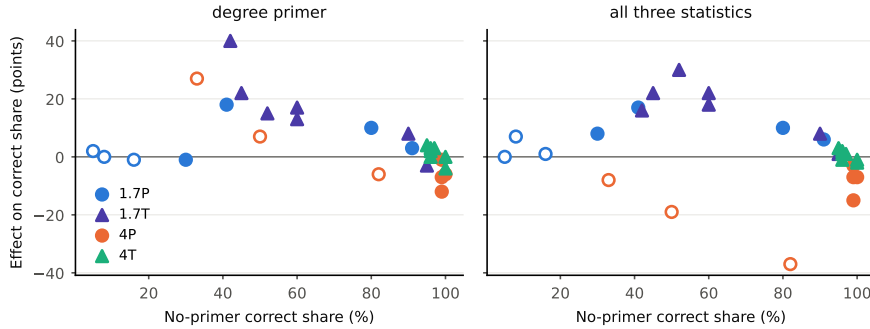


Figure 1: Primers that state the answer, on node degree: effect (points) against the no-primer correct share, per model and density. Left: degree; right: all three statistics (the same degrees in a longer primer). Hollow: at least 15% truncation, or the plain models’ dense runs.

cal clustering (a normalized triangle count) and random-walk return probabilities (RWSE)—stated as sentences. A graph-blind solver answers from the primer alone. Because added text and order change answers, we fix one layout, compare every primer with both no primer and a structure-free *filler* about as long as the clustering primer, and check what responses claim about the graph. Qwen3 runs the same weights with and without thinking (Qwen Team, 2025). A pilot on the published GraphQA graphs (5–19 nodes) with Gemma 4 and Qwen3-8B/14B found most arms near ceiling without a primer (Appendix C), so the main study uses larger graphs and Qwen3-1.7B and Qwen3-4B, each in plain and thinking mode (four

arms; P/T below), zero-shot.

Graphs, tasks and primers. Graphs are undirected Erdős–Rényi graphs (Gilbert, 1959) with 40 nodes at edge probabilities $p \in \{.10, .20, .35, .50\}$ (the *main sweep*), and for node degree and edge existence also $\{.65, .75, .85\}$ (the *dense extension*), with 100 graphs per task and density shared by every condition and model; decoding is greedy, one response per prompt, capped at 8,192 tokens (2,048 for the plain models’ dense runs). The incident encoding lists each node’s neighbors on its own line (“Node 3 is connected to nodes 1, 7, 12.”). We use GraphQA’s six tasks: node degree, connected nodes (the queried node’s neighbors), edge count,

edge existence, node count and cycle check. At 40 nodes the node count is always 40 and every graph has a cycle, so those two tasks show how a prompt changes the answer rather than how the graph is read. Before the encoding we place no primer, a *degree*, *clustering* or *RWSE* primer (return probabilities after 2 and 3 steps), or *all* three, one sentence per node with values to two decimals; two diagnostics are a sentence stating the number of connected components and the filler, which names every node without any structure (“Node 0 is simply present within the graph G.”; details in Appendix A).

Outcomes and analysis. Each response is **correct**, **wrong** or **truncated** (it used the whole budget); these three shares have all prompts in their denominator, shares of finished answers are marked as such, and an effect is the paired change in the correct share on the same graphs (Table 1). A cell with at least 15% truncation on either side is *flagged*. Every task is scored by exact match, as in GraphQA (for connected nodes, the whole neighbor set); edge existence is also scored by balanced accuracy (Brodersen et al., 2010), because the share of true edges rises from 10% to 85% with density. Paired comparisons use exact McNemar tests (McNemar, 1947), bootstrap intervals over graphs and Benjamini–Hochberg correction within each (model, task, control) family (Benjamini and Hochberg, 1995); q is the adjusted p , and we call an effect significant at $q < .05$. The main sweep and dense extension comprise 84,000 responses on 700 graphs.

Follow-up runs. Node degree is re-tested with 400 graphs per density for plain and thinking Qwen3-1.7B (no primer, components, clustering, degree, filler), on a fresh seed (1,600 graphs), and on a grid of 20 to 160 nodes at mean degree 8 and 16 (400 graphs per cell, also Qwen3-8B).

4 Results

The graph-blind solver answers node degree and edge count perfectly from the degree primer or all three statistics, by reading the queried degree or halving the sum of the stated degrees; among the other primers and tasks whose answer varies it never exceeds 74.6%, against 73.5% with no primer. These answer-stating cells are shaded in Table 1; we call every other primer–task pair side information.

4.1 Stating the answer

A stated answer helps where the model computes it unreliably: the degree primer raises node-degree accuracy for plain and thinking Qwen3-1.7B by 7.5 and 8.8 points ($q = .0042, .0008$; truncation +4.5 for the latter) but costs plain Qwen3-4B, which already counts correctly, 6.5 ($q < .001$). The gain peaks at intermediate accuracy without the primer (Figure 1): over all unflagged cells of the four tasks whose answer varies, answer-stating primers gain 8.3, 16.1 and 12.9 points where that accuracy is below 25%, 25–50% and 50–75%, and lose 5.8 at 75–90% (split-half +13.9 and +16.2, so not regression to the mean). One model can fall on either side: the degree primer costs plain Qwen3-4B 12.0 points at $p = .50$, where it answers 99% correctly, and gains 27.0 at $p = .85$, where it answers 33% ($q = .0064, .0021$). A model that has stopped reading the graph is not rescued: at $p = .85$ plain Qwen3-1.7B answers 39 (all other nodes) on 70.8% of finished answers even with the degree stated. On edge count the degree primer supplies the terms of the sum: plain Qwen3-4B rises from 2.00% to 29.25% (+27.2), the largest gain in Table 1 on a task whose answer varies.

4.2 Changing the procedure

A stated value replaces the model’s own count or checks it. Plain Qwen3-4B enumerates the queried node’s neighbors in every response at $p = .50$ without a primer; with it, 62% of responses read the value off (a *look-up*), and 13 answers break against 1 fixed; over all densities, 17 of its 57 wrong look-ups equal the degree stated just above or below the queried node’s, against 10.4 expected by chance ($p = .017$). Look-ups fail more often in a longer primer: inside all three statistics, the degrees cost dense plain Qwen3-4B 21.3 points ($q < .001$) where the degree primer alone gains 9.3 ($q = .03$; all three against degree alone, $-30.7, q < .001$), and over all densities its look-ups under all three are right for 88.5% of queried nodes 0–9 but 53.4% of nodes 10–39 (within-density permutation $p = .0005$). Thinking models that report a conflict between the stated value and their own count end on the stated value in 93% (4B) and 66% (1.7B) of cases, while plain Qwen3-1.7B gains without quoting the value (0 of its 39 fixes at $p = .20$ and $.35$ cite it). A sentence that states no answer can change the procedure too: the component sentence, the same for every connected graph, raises plain

Qwen3-4B’s neighbor sets by 9.2 ($q < .001$), and with it the model restates the queried node’s line in 98.5% of responses, against 63.0%.

4.3 Side information

Side information is read against what any added text does. Filler adds no information, yet it flips largely the same questions as the primers: a question filler flips is also flipped by a primer 38.5–69.1% of the time, against 3.1–15.2% of other questions (pooled over tasks; permutation $p = .0005$ in all 20 tests). Filler significantly lowers accuracy in four cells: plain Qwen3-1.7B’s edge existence (−15.8 points, balanced accuracy −10.8) and neighbor sets (−6.2), and plain Qwen3-4B’s edge existence (−7.0, −4.5) and neighbor sets (−12.5). In the first and last, clustering and RWSE are 19.5 and 20.8 points above filler, and 13.5 and 12.5 (all $q < .001$; balanced accuracy on the first +13.3 and +13.6, $q < .001$). Components, clustering and RWSE add little on average: where accuracy without a primer is 25–75%, they gain 1.6 points while filler loses 6.9 in the same cells, so they are 8.5 points above filler there. On plain Qwen3-4B’s edge existence clustering gains 4.0 points and RWSE loses 10.0 (balanced accuracy +3.0 and −6.5, $q = .018$ and $.0015$); on its dense node-degree queries clustering gains 11.3 ($q < .001$). The clearest benefit is in what plain Qwen3-1.7B claims: with a statistic it rarely states a queried non-edge as an edge (13.7% → 2.7–4.4%; $p \leq .50$, finished answers), and all three statistics lower its *false alarms*, yes answers on non-edges, from 0.51 to 0.33 over all densities (finished answers) and raise its balanced accuracy by 7.8 points at $p \leq .50$ ($q = .001$) and by 14.9 at $p \geq .65$, where without a primer it answers yes to nearly every pair ($q = .0025$).

A replicated effect. Clustering on node degree for plain Qwen3-1.7B (+3.0, $q = .38$ in Table 1) is the one side-information effect a 40-cell screen selected for re-testing (Figure 2; Appendix C). At 40 nodes and $p \leq .50$ it gains 4.0 points on new graphs ($p = .0047$), 4.2 on a fresh seed ($p = .00055$) and 3.2 against filler ($q = .014$), and 6.2 over 20–160 nodes at mean degree 8 and 16 ($p < .001$). At $p \geq .65$ it is absent (−0.7; density difference $q = .0044$), for thinking Qwen3-1.7B there it is a loss (−5.4, $q = .0015$), and for Qwen3-8B on the grid it is absent (−1.0). It mostly corrects errors in copying the neighbor list (224

Arm	Primer	Correct	Valid	False		No cycle	
		“yes” (<i>n</i>)	Real cycle	Invented cycle	Not a cycle	All	Edge count
1.7P	none	363	36.9	43.5	16.0	3.6	0.8
	clustering	397	39.5	48.1	7.8	4.5	1.0
	RWSE	362	33.7	40.3	19.9	6.1	1.7
	degree	293	13.0	20.5	1.4	65.2	55.6
	all	391	38.4	50.4	7.9	3.3	1.3
1.7T	none	388	79.9	17.8	1.0	1.3	0.5
	clustering	386	80.6	19.2	0.0	0.3	0.3
	RWSE	397	68.0	20.7	10.6	0.8	0.5
	degree	301	61.5	29.6	0.0	9.0	9.0
	all	385	73.0	21.8	3.1	2.1	1.8
4P	none	396	34.1	23.0	6.8	36.1	25.0
	clustering	398	48.7	31.2	1.0	19.1	5.8
	RWSE	395	33.2	20.8	38.7	7.3	0.5
	degree	387	13.4	3.9	2.1	80.6	65.6
	all	395	45.1	33.4	6.6	14.9	0.8
4T	none	397	96.5	3.3	0.0	0.3	0.0
	clustering	397	95.5	4.5	0.0	0.0	0.0
	RWSE	398	55.0	2.5	42.5	0.0	0.0
	degree	375	90.9	5.1	0.0	4.0	2.7
	all	395	92.7	6.3	0.0	1.0	1.0

Table 2: What the correct cycle-check answers rest on (main sweep; every correct answer is “yes”). *Correct* (*n*): count of correct, finished answers; the other columns are % of those *n*, and Real, Invented, Not a cycle and All sum to 100. *Valid*: a cycle whose every step is an edge. *False*: *Invented* has a step that is not an edge; *Not a cycle* is a length-2 or retraced walk. *No cycle*: *All* names none; *Edge count* argues from the number of edges (valid). Bold (Real, Invented only): differs from the same model without a primer (paired, BH $q < .05$).

corrected, 176 introduced, all densities) without any response mentioning clustering.

4.4 Correct is not executed

Every graph has a cycle, so every correct cycle-check answer is “yes”; Table 2 checks, edge by edge, what it rests on. Without a primer only 36.9% of plain Qwen3-1.7B’s and 34.1% of plain Qwen3-4B’s rest on a real cycle, and 43.5% of plain Qwen3-1.7B’s on an invented one. With the degree primer the plain models argue from the number of edges instead, a valid argument (55.6% and 65.6%), though the primer costs plain and thinking Qwen3-1.7B 17.5 and 21.8 points (truncation +8.2 and +21.2) and thinking Qwen3-4B 5.5 ($q < .001$). Clustering and all three statistics make plain Qwen3-4B rest on a real cycle more often (+14.7, +11.5) but on an invented one too (+8.4, +10.0; $q < .05$), and RWSE leads thinking Qwen3-4B to rest 42.5% of its correct answers on a walk that is not a cycle (length 2 or retraced).

4.5 Thinking and the budget

Thinking improves execution. It raises the share of correct cycle-check answers resting on a real cycle to 80.9% and 96.4% on graphs both modes answer correctly ($q < .05$), and on node degree it adds more than any statistic that does not state the answer: 24.8 points for Qwen3-1.7B without a primer in the follow-ups, from +1.5 at $p = .10$ to +43.5 at $p = .85$, and 35.5 with the degree stated, at 6.2–17.1 times the plain mode’s median tokens. Thinking Qwen3-4B, near ceiling, moves by at most 2.5 under any primer on node degree, edge existence and neighbor sets. On edge count it is right on 97.6% of the answers it finishes but runs out of budget on 79.5% of prompts; the component sentence’s gain there (+24.8) is a fall in truncation (−25.0; Appendix Figure 3).

5 Discussion

Substitution, not improvement. The largest gains come from stating the answer where the model computes it unreliably, and a model that already counts correctly loses by switching to a look-up; one model does both as density changes. A stated answer patches weak execution rather than improving it, replacing a count with a look-up that has its own errors. Whether to add one therefore depends on how well the model computes the quantity without it.

Findable content. More statistics are not better on average: all three together are no better than the mean of their parts (+0.3 over 53 combinations). Dense plain Qwen3-4B loses the degrees in the longer primer, and rounded RWSE takes only 3.30 distinct value pairs per graph at $p = .50$ against 12.50 distinct degrees, so it stops telling nodes apart as density rises.

Side information acts on claims. A statistic that does not state the answer acts most clearly on what the model claims, cutting the non-edges plain Qwen3-1.7B states where structure-free text adds them (Appendix B). Because filler flips largely the same questions as the primers, part of any primer’s effect is a response to added text, so each effect should be read against both controls.

Measuring execution. Raw edge-existence accuracy rewards saying yes on dense graphs: plain Qwen3-1.7B’s rises from 54% to 86% between $p = .50$ and $.85$ as its balanced accuracy falls to

0.53. Truncation is its own outcome, and a correct answer need not rest on the graph: as with unfaithful chain-of-thought (Turpin et al., 2023) and pattern-matched composition (Dziri et al., 2023), a right label is weak evidence of execution. Graph benchmarks can report balanced accuracy, truncation and what correct answers rest on.

Thinking and statistics. Thinking improves execution itself: it adds more than any side statistic on Qwen3-1.7B’s node degree and raises the share of correct cycle answers resting on a real cycle from about a third to 80.9–96.4%, at a cost in tokens.

6 Future Work

Three controlled tests follow: primers that withhold the queried node’s own value; randomized node labels and preamble order against token-matched neutral text; and variable-size, acyclic and non-Erdős–Rényi corpora, scored also for consistency across queries. More broadly, the results invite comparing precomputed statistics with graph tools and model-written code; predicting a primer’s benefit from what it reveals, the model’s shortfall and its reading cost; summarizing graphs too large for the encoding; searching over feature, wording and placement; testing whether models verify corrupted statistics; locating a primer’s effect with activation patching; and releasing these controls as a benchmark.

7 Conclusion

Mostly no: explicit structural statistics raise accuracy mainly by supplying the answer, which substitutes for execution rather than improving it, helping where a model computes it unreliably and costing where it computes it well. Other statistics have small, model-specific effects on average and act most clearly on what a model claims; one, clustering on node degree for plain Qwen3-1.7B, replicates. Thinking does improve execution. Because correct answers can rest on invented structure, evaluations of LLMs on graphs should check what an answer rests on, not only whether it is right.

Limitations

We study one model family (Qwen3), Erdős–Rényi graphs at several densities (40 nodes; 20–160 in one follow-up), one encoding and one wording per primer, zero-shot. Node ids 0–9 are both first and single-digit, so node-level results are stated by id.

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A Method Details and Audit Trail

Statistics. With adjacency matrix A and transition matrix $P = D^{-1}A$, the per-node statistics are $d_v = \sum_u A_{vu}$, $c_v = 2t_v/(d_v(d_v - 1))$ and $r_v^{(k)} = (P^k)_{vv}$ for $k \in \{2, 3\}$, where t_v counts edges among v ’s neighbors and $c_v = 0$ when $d_v < 2$. Clustering and return probabilities are printed to two decimals, one sentence per node, in node order.

Graph-blind solver. It uses 16 exact rules, 1 heuristic and 8 rules fitted on separate graphs.

Why these features. Degree and RWSE are the structural encodings graph Transformers use (Ram-pásek et al., 2022); local clustering adds how a node’s neighbors connect; the component count is a single global fact that, with the node and edge counts, decides whether a cycle exists. Laplacian encodings are left out because their sign and basis are arbitrary and cannot be stated as facts (Dwivedi et al., 2022).

Detectable effects. In one cell of 100 graphs the median smallest detectable paired effect is 7.4 points; among contrasts without a significant effect (400 paired graphs each), the median detectable gain is 5.7 points for the 1.7B models and 3.2 for the 4B models.

Response-level measures. Rule-based readers extract what each response states: whether it enumerates the queried node’s neighbors or retrieves a stated value, whether it reports a conflict with the primer, and which cycles and edges it asserts, each checked edge by edge against the prompt’s graph.

Coverage. Every result uses an undirected graph with exactly 40 nodes, except the fixed-mean-degree grid (20–160 nodes) and the pilot; in the main sweep each model sees the same graph instances under seven conditions, so pairs are formed on (task, instance, model). Table 3 gives the primer lengths, and Appendix D the complete tables.

Scoring and provenance. A generation that reaches its output cap is *truncated*. Set-F1 on connected nodes is a supplement, scoring a truncated response as zero; changes in balanced accuracy use a paired permutation test with Benjamini–Hochberg adjustment within each model; cells with at least 15% truncation on either side are flagged and left out of pooled summaries. Tagged results come from GraphTalk’s `results-sot` branch

Condition	Mean chars	Information
Components	37	Number of connected components
Degree	891	Degree for each node
Clustering	1,629	Local clustering for each node
RWSE	2,949	Two rounded return values per node
All three	4,691	Degree, clustering and RWSE
Filler	1,829	Node names, no graph statistic

Table 3: Mean preamble length in characters over the main-sweep graphs.

(GraphTalk Contributors, 2026): each tag names a block of a script output cited by the results documents in `docs/`.

B Additional Results

Figure 2 shows the clustering primer on node degree for every model and follow-up (§4), and Figure 3 thinking Qwen3-4B’s edge-count outcomes by condition.

Presentation effects. Under filler, a preamble that adds no information, plain Qwen3-1.7B largely stops listing node ids on node count (59.5% → 12.2% of answers), states more non-existent edges (+9.6), plain Qwen3-4B invents more neighbors (+14.8), and thinking Qwen3-4B runs out of budget less often on edge count (78.2% → 65.0%)

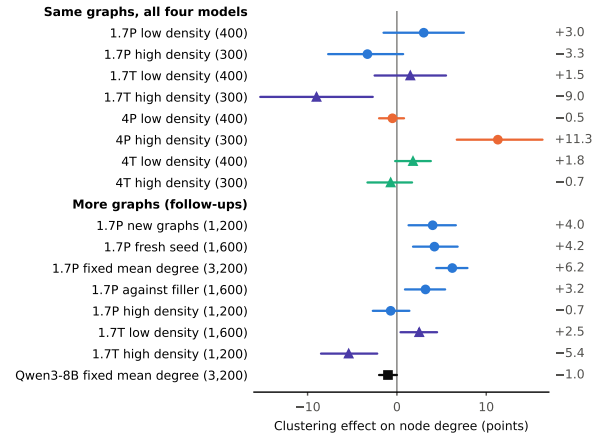


Figure 2: The clustering primer on node degree: effect in points with 95% bootstrap intervals; graphs per row in brackets. Top: all four models on the main-sweep graphs at low ($p \leq .50$) and high density ($p \geq .65$). Bottom: follow-ups with more graphs for plain Qwen3-1.7B, where a screen found the effect, its thinking mode, and Qwen3-8B (black squares) on the fixed-mean-degree grid ($n = 20$ –160).

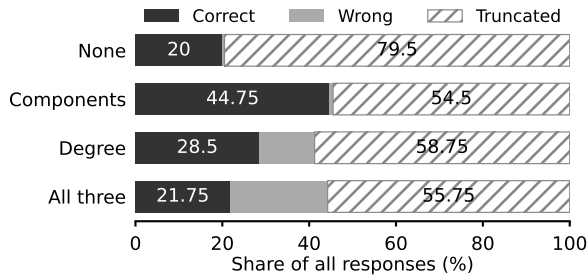


Figure 3: Thinking Qwen3-4B on edge count: correct, wrong and truncated shares of *all* responses. Conditions from top: none, component count, degree, all three.

truncated while working), matching the sensitivity of LLM reasoning to added and irrelevant text (Levy et al., 2024; Shi et al., 2023).

Clustering by node id. The replicated clustering gain for plain Qwen3-1.7B falls on queried nodes with low ids (+5.01, +0.75 and -0.11 points by thirds over all seven densities; $p = .018$ for the trend), which filler does not show.

Reading errors and thinking. On node count, whose answer is always 40, plain Qwen3-1.7B answers 39, the last id rather than a count, on 68.5% of finished responses without a primer, 1.2% under clustering and 28.7% under filler. Without a primer, plain Qwen3-1.7B’s stated neighbor lists on node degree miss a neighbor in 24.6% of cases and invent one in 3.3%. Plain Qwen3-1.7B’s edge-existence false alarms rise with the neighbors the two nodes share, from 1.7% of non-edges with none shared to 65.9% with four or more ($p \leq .50$, all conditions pooled). For plain Qwen3-4B, clustering is the only primer that lowers its false alarms ($0.16 \rightarrow 0.08$, all densities, finished). On neighbor sets, 93.0% of plain Qwen3-4B’s missed neighbors lie below the queried node’s id, against 43% by chance (all conditions pooled). Under the degree primer, thinking Qwen3-1.7B’s first candidate answer appears earlier in its trace (median 27% of the way through, against 44%). Plain models run out of budget in loops (83.9% and 89.6% of their truncated responses), thinking models while still working.

C Screening and Focused Robustness Checks

Earlier screens. A pilot on the published GraphQA graphs (5–19 nodes) evaluated Gemma 4 E4B/12B and Qwen3-8B/14B, each in plain and

Arm / primer	Low	High	High-low	q
1.7P / cluster.	+3.81	-0.67	-4.48	.0044
1.7T / cluster.	+2.50	-5.42	-7.92	< .001
1.7T / degree	+11.31	+24.75	+13.44	< .001

Table 4: Primer effects on degree correct share (points) relative to no primer. Low: $p \leq .50$ (1,600 pairs); high: $p \geq .65$ (1,200 pairs). P/T: plain/thinking.

thinking mode; most arms were near ceiling without a primer on most tasks.

Selection and replication. The clustering-on-degree condition emerged from a 40-cell screen: 700 dedicated graphs overlap it, while 2,100 do not. The independent fresh seed gives +4.2 points ($p = .00055$), with screen-wide Bonferroni-adjusted $p = .022$.

Sensitivity. Split-half means remain +13.9 and +16.2 points for answer-carrying cells at intermediate baselines, and the band means hold when the truncation flag is set at 10% or 50%.

Fixed-size effect heterogeneity. Table 4 compares primer effects between density groups; `robustness_audit.py` regenerates it from saved responses.

Direct bundle contrast. Across 300 paired dense plain-4B degree queries, clustering alone is correct on 66.33% versus 33.67% with *all* (-32.67 points, $q < .001$ across six alternatives).

D Complete Tables

Table 5 gives the main sweep’s secondary metrics: set-F1 on neighbor sets, and raw and balanced accuracy, yes-rate and truncation for every edge-existence condition. Table 6 gives the truncated share for every cell of Table 1, so each cell’s wrong share is the remainder. Table 7 repeats the comparison on the dense extension for node degree and edge existence.

(a) Neighbor-set Set-F1					(b) Edge-existence decisions				
Arm	Primer	None	With	Δ ; q	Arm	Primer	Raw	Bal.	Yes Trunc.
1.7P	filler	0.9721	0.9624	-0.0096; 0.3114	1.7P	none	70.50	79.86	0.562 0.000
1.7P	components	0.9721	0.9716	-0.0004; 0.9503	1.7P	filler	54.75	69.11	0.720 0.000
1.7P	clustering	0.9721	0.9724	+0.0003; 0.9503	1.7P	components	62.25	74.23	0.645 0.000
1.7P	rwse	0.9721	0.9702	-0.0018; 0.9503	1.7P	clustering	74.25	82.42	0.525 0.000
1.7P	degree	0.9721	0.9794	+0.0074; 0.3352	1.7P	rwse	75.50	82.68	0.501 0.003
1.7P	all	0.9721	0.9626	-0.0094; 0.3846	1.7P	degree	74.50	82.30	0.517 0.000
1.7T	filler	0.9732	0.9769	+0.0037; 0.8851	1.7P	all	82.00	87.71	0.448 0.000
1.7T	components	0.9732	0.9726	-0.0006; 0.9942	1.7T	none	99.00	99.02	0.271 0.003
1.7T	clustering	0.9732	0.9830	+0.0098; 0.4254	1.7T	filler	98.50	98.68	0.278 0.000
1.7T	rwse	0.9732	0.9577	-0.0156; 0.4254	1.7T	components	98.25	98.81	0.283 0.003
1.7T	degree	0.9732	0.9627	-0.0105; 0.4254	1.7T	clustering	98.00	98.63	0.286 0.003
1.7T	all	0.9732	0.9733	+0.0001; 0.9942	1.7T	rwse	97.75	98.17	0.283 0.003
4P	filler	0.9871	0.9660	-0.0212; 0.0001	1.7T	degree	96.50	97.61	0.299 0.005
4P	components	0.9871	0.9976	+0.0105; 0.0001	1.7T	all	97.25	98.12	0.295 0.000
4P	clustering	0.9871	0.9845	-0.0026; 0.5411	4P	none	90.50	93.22	0.357 0.000
4P	rwse	0.9871	0.9776	-0.0095; 0.0681	4P	filler	83.50	88.74	0.432 0.000
4P	degree	0.9871	0.9768	-0.0103; 0.0379	4P	components	87.75	91.64	0.390 0.000
4P	all	0.9871	0.9743	-0.0128; 0.0209	4P	clustering	94.50	96.25	0.323 0.000
4T	filler	0.9636	0.9720	+0.0084; 0.8279	4P	rwse	80.50	86.69	0.463 0.000
4T	components	0.9636	0.9641	+0.0006; 0.8956	4P	degree	85.75	90.27	0.407 0.005
4T	clustering	0.9636	0.9544	-0.0092; 0.8279	4P	all	80.50	86.69	0.461 0.003
4T	rwse	0.9636	0.9676	+0.0041; 0.8279	4T	none	99.75	99.83	0.268 0.003
4T	degree	0.9636	0.9685	+0.0050; 0.8279	4T	filler	99.75	99.83	0.268 0.003
4T	all	0.9636	0.9551	-0.0084; 0.8279	4T	components	99.25	99.49	0.270 0.007
					4T	clustering	99.00	99.32	0.270 0.010
					4T	rwse	98.50	98.68	0.269 0.015
					4T	degree	99.75	99.83	0.268 0.003
					4T	all	98.75	99.15	0.271 0.013

Table 5: Secondary metrics on the same 400 main-sweep prompts per model and condition. (a) Neighbor-set set-F1, with truncations scored zero: each row gives the no-primer baseline, the primer score and paired difference; q adjusts six contrasts per model (paired random-sign tests). (b) Edge-existence raw and balanced accuracy (Bal., %) and truncated share (Trunc.) use all prompts; yes-rate (Yes) uses finished answers. Compare each row with its model’s none row; changes in balanced accuracy are tested in the text (§4). P/T: plain/thinking.

Task	Arm	None	Degree	Cluster.	RWSE	All	Comp.	Filler
Degree	1.7P	0.00	+0.0	+0.0	+0.2	+0.0	+0.0	+0.0
	1.7T	0.75	+4.5	-0.5	+0.0	-0.5	-0.2	-0.5
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	2.75	-2.5	-2.0	-0.5	-1.0	-0.5	-1.8
Neighbors	1.7P	0.00	+0.0	+0.0	+0.0	+0.2	+0.0	+0.0
	1.7T	1.25	+0.2	-1.0	-0.8	-0.5	-0.2	-0.8
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	3.25	-0.2	+1.2	-0.2	+1.0	+0.2	-0.5
Edge count	1.7P	29.75	-20.2	-11.8	-11.8	-1.8	-2.5	-3.5
	1.7T	81.75	-0.8	+0.8	+5.0	-12.2	+0.5	+1.2
	4P	0.00	+1.8	+0.2	+0.5	+0.8	+0.8	+0.0
	4T	79.50	-20.8	-26.0	-12.2	-23.8	-25.0	-13.0
Edge exists	1.7P	0.00	+0.0	+0.0	+0.2	+0.0	+0.0	+0.0
	1.7T	0.25	+0.2	+0.0	+0.0	-0.2	+0.0	-0.2
	4P	0.00	+0.5	+0.0	+0.0	+0.2	+0.0	+0.0
	4T	0.25	+0.0	+0.8	+1.2	+1.0	+0.5	+0.0
Node count	1.7P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	1.7T	7.75	-5.0	-7.8	-7.8	-7.8	-0.5	-7.8
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	0.00	+0.2	+0.2	+0.2	+0.0	+0.0	+0.0
Cycle	1.7P	9.00	+8.2	-8.2	-7.2	-8.0	-2.5	-2.8
	1.7T	3.00	+21.2	+0.5	-2.2	+0.8	+7.5	-3.0
	4P	0.75	+0.8	-0.2	+0.5	-0.2	+0.8	+0.5
	4T	0.75	+4.8	+0.0	-0.2	+0.5	+3.2	-0.5

Table 6: Complete main-sweep truncated shares: no-primer baseline (percent of all prompts) and paired changes (percentage points), for the same queries as Table 1. Each of its cells’ *wrong* share equals 100% minus its correct and truncated shares. P/T: plain/thinking; components and filler are diagnostics. No significance marks are assigned to truncation contrasts.

(a) Correct share								
Task	Arm	None	Degree	Cluster.	RWSE	All	Comp.	Filler
Degree	1.7P	9.67	+0.3	-3.3	-5.3	+2.7	+0.3	-5.3
	1.7T	46.33	+25.7	-9.0	-4.3	+22.7	-1.7	-12.0
	4P	55.00	+9.3	+11.3	+7.7	-21.3	-1.0	+2.0
	4T	97.33	-0.7	-0.7	-0.7	+0.3	-2.0	+0.0
Edge exists	1.7P	77.00	+2.0	+1.3	+1.0	+6.7	-1.0	-1.3
	1.7T	99.00	-0.7	-0.3	+0.3	-0.7	+0.0	-0.7
	4P	90.00	-3.3	+5.3	-0.7	-2.3	-3.3	-6.3
	4T	99.33	+0.7	+0.7	+0.3	+0.3	+0.7	+0.7

(b) Truncated share								
Task	Arm	None	Degree	Cluster.	RWSE	All	Comp.	Filler
Degree	1.7P	0.33	-0.3	-0.3	-0.3	-0.3	-0.3	-0.3
	1.7T	1.67	+9.0	-1.3	-1.3	+0.0	+0.3	-1.0
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	0.33	+0.3	+0.0	+0.7	+0.7	+0.3	-0.3
Edge exists	1.7P	0.00	+1.3	+0.0	+0.0	+0.0	+0.0	+0.0
	1.7T	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4P	0.00	+0.3	+0.0	+0.0	+0.3	+0.3	+0.0
	4T	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0

(c) Edge-existence decisions								
Measure	Arm	None	Degree	Cluster.	RWSE	All	Comp.	Filler
Balanced acc.	1.7P	52.79	+8.6	+3.9	+2.1	+14.9	-2.1	-2.8
	1.7T	99.31	-1.7	-0.4	-1.1	-1.2	+0.0	-1.3
	4P	83.68	-9.3	+9.5	-3.6	-9.7	-7.3	-13.5
	4T	99.61	+0.4	+0.4	+0.2	-0.7	+0.4	+0.4
Yes-rate	1.7P	98.67	-6.2	-2.7	-1.0	-8.0	+1.0	+1.3
	1.7T	74.67	+0.7	+0.3	+1.7	+1.3	+0.0	+0.0
	4P	80.33	+6.6	-4.0	+1.3	+6.3	+4.6	+7.0
	4T	75.00	+0.7	+0.7	+0.3	+1.0	+0.7	+0.7

Table 7: Complete dense-extension comparison at $p \in \{.65, .75, .85\}$ for degree and edge existence: no-primer value (%) and paired change (points). (a,b) Correct and truncated share of all 300 prompts; the wrong share is the remainder. (c) Balanced accuracy (a truncated response is not correct) and yes-rate of finished answers, each averaged over the three levels. The contrasts cited in the text are tested; the other cells are descriptive. The plain models’ budget here is 2,048 tokens. P/T: plain/thinking.